

Helix OTA — AGENTS.md

Base agent rules live at constitution/AGENTS.md and the constitution/Constitution.md it references. **READ THOSE FIRST.** The base file is authoritative for any topic not covered here. This file extends them with project-specific rules; it never weakens them.

Critical base rules restated (for agents that don't follow @imports)

- **No bluffing.** Every PASS carries positive evidence. Constitution §11.4.
- **Mutation-paired gates.** Every new gate has a paired mutation proving it catches regressions. Constitution §1.1.
- **No guessing language.** likely, probably, maybe, seems, appears etc. are forbidden when reporting causes. Constitution §11.4.6.
- **Credentials never tracked.** .env patterns git-ignored; runtime-load only; per-service file separation. Constitution §11.4.10.
- **Never force-push.** Force-push requires explicit per-session authorization. Constitution §11.4.113.
- **Hardlinked backup before any destructive op.** Constitution §9.
- **CONTINUATION document kept in sync.** Constitution §12.10. The live handoff for this project is docs/research/main_specs/CONTINUATION.md (and .remember/remember.md when present).
- **60% RAM cap.** Heavy work wrapped in bounded execution scope. Constitution §12.6.
- **Per-feature Status + video confirmation.** Constitution §11.4.153. docs/features/Status.md + Status_Summary.md enumerate every feature with per-row video-recording confirmation; HTML+PDF+DOCX exports in sync.
- **Window-scoped capture + fresh-corpus rotation.** Constitution §11.4.154. Recordings capture only the app window/surface, not whole desktop; a new run removes its own prior in-scope recordings first.
- **Project-prefixed recording filenames.** Constitution §11.4.155. Every recording filename starts with <PREFIX>--- (resolved from

HELIX_RELEASE_PREFIX env var or project-root dir name), not a single hyphen.

- **All CI/CD disabled.** Constitution §11.4.156. No GitHub Actions or GitLab pipelines active; enforcement is local-only via pre-build gates and git hooks.
- **GEMINI.md lockstep.** Constitution §11.4.157. GEMINI.md kept in sync with CLAUDE.md/AGENTS.md/QWEN.md at all times.
- **Intensive recording + read-the-screen content verification.** Constitution §11.4.158. Every feature/flow/edge-case recorded; every recording's on-screen content machine-read and verified as a genuine working result.
- **Window-specific MP4 + vision validation.** §11.4.159. Every recording must be window-specific MP4 with expected-content specification BEFORE recording, vision validation AFTER recording, terminal window cleanup, and content-verification workflow.
- **Vision-verified recording + HelixQA bridge.** Constitution §11.4.160. Recordings processed through a vision/OCR bridge confirming expected results before acceptance.
- **Rootless containers.** Constitution §11.4.161. The containers/ infra runs Podman rootless via the vasic-digital/containers submodule; no Docker-rootful / sudo / root escalation.
- **OpenDesign UI system.** Constitution §11.4.162. Any user-facing UI uses OpenDesign design tokens (light+dark, visual-regression covered); latent until a UI surface ships.
- **Media validation pipeline.** Constitution §11.4.163. Every recorded artifact validated (OCR / transcription / parse vs SPECIFY-phase patterns, self-validated analyzer) before acceptance.
- **Constitution auto-propagation hook.** Constitution §11.4.164. After every constitution pull run constitution/scripts/post_update_hook.sh (inherited by reference) to register/install changed skills/MCP/hooks.
- **Independent verification agent.** Constitution §11.4.165. Every code change OR media artifact passes an independent verifier iterating to a zero-finding GO.
- **Semgrep static analysis.** Constitution §11.4.166. Semgrep installed + on PATH; semgrep scan --config auto --error runs before every commit / push and blocks on findings; scripts inherited by reference from constitution/scripts/semgrep/*.
- **Big-work-item feature work-stream lifecycle.** Constitution §11.4.167. Every BIG feature/large fix develops as its own isolated feature work-stream (CoW/reflink project copy, own feature/<slug> branch + tags, own builds), separate from trunk until operator-approved after full retest; trunk merged in regularly; submodule

branch/tag cascade (composes §11.4.58/.103/.113/.142/.145). Gate CM-COVENANT-114-167-PROPAGATION.

- **Exported-document independent visual validation.** Constitution §11.4.168. Every exported document (HTML/PDF/DOCX) passes independent validation (reviewer structurally separate from the generator) across CONTENT, TEXTUAL, and FULL-VISUAL layers, verified by rendering + OCR with a self-validated golden-good/golden-bad analyzer (composes §11.4.65/.107/.117/.134/.142). Gate CM-COVENANT-114-168-PROPAGATION.
- **Mandatory comprehensive test-type coverage.** Constitution §11.4.169. Every project covers the closed enumerated test-type set (unit/ integration/e2e/full-automation/Challenges/HelixQA/DDoS/ security/ stress+chaos/concurrency/race-deadlock/memory/ benchmarking), each PASS citing rock-solid captured physical evidence (composes §11.4.25/.27/ .50/.52/.69/.85/.107). Gate CM-COVENANT-114-169-PROPAGATION.
- **Host-rendered UI visual-proof mandate.** Constitution §11.4.170. Every UI-surface change is proven by device-independent host-rendered pixels (Compose/Playwright/Storybook/snapshot-testing class) per screen×state×{light,dark} theme, validated by golden image-diff AND an OCR/vision layout oracle; value/token-equality unit tests are FORBIDDEN as the sole proof (composes §11.4.107/.153/.158/.159/.160/.162/.168). Gate CM-COVENANT-114-170-PROPAGATION.
- **Human-readable workable-item descriptions.** Constitution §11.4.171. Every workable item carries a ≥5-7-sentence plain-language description (what/why/how/who-benefits/expected-outcome) understandable by non-developers; the SQLite DB description column is the single source of truth mirrored into every derived doc (composes §11.4.15/.16/ .91/.93/.148). Gate CM-COVENANT-114-171-PROPAGATION.
- **Production-readiness planning mandate.** Constitution §11.4.172. Every project maintains a living planning document with realistic timeline projections from measured velocity, danger-zone/risk identification, and critical-path analysis, updated monthly or on ≥10% item-count change (composes §11.4.6/.40/.42/.93/.108). Gate CM-COVENANT-114-172-PROPAGATION.
- **Containerized + distributed build mandate.** Constitution §11.4.173. Every build of every component runs inside a specialized build container via the containers submodule, distributed to the designated remote build host (never the bare host), artifacts brought back for use/flashing (composes §11.4.28/.74/.76/.108/.161). Gate CM-COVENANT-114-173-PROPAGATION.

- **Shared-host process-ownership verification.** Constitution §11.4.174. Before inspecting or acting on any process/build/daemon/port/lock on a shared host, positively verify the target is OURS (cwd/argv/ recorded-PID/lock-path) — never a loose name-match pgrep; never kill a process not positively ours (composes §11.4.66/.101/.133/.147). Gate CM-COVENANT-114-174-PROPAGATION.
- **Multi-track work-division + device-lock arbitration.** Constitution §11.4.176. (A) exactly-once work-item/logical-group claim registry; 1. capability-aware deadlock-proof device-lock (all-or-nothing, non-blocking, TTL-reap breaking Coffman conditions); (C) universal/ decoupled + evidence-based resource tuning (composes §11.4.58/.116/.119/.147/.167). Gate CM-COVENANT-114-176-PROPAGATION.
- **Developer-tooling project-decoupling.** Constitution §11.4.177. No project-specific script/hook/alias may be wired into a global/shared developer-tooling PATH; shared tooling is project-agnostic and operates on the invocation directory, never a hardcoded project path (composes §11.4.28/.29/.35). Gate CM-COVENANT-114-177-PROPAGATION.
- **Track-qualified identity for parallel streams.** Constitution §11.4.178. Parallel work streams sharing hardware/git/session/lock namespaces are addressed by a track-qualified identity (<project>__<track>__<role>), never a bare basename — prevents session/lock/log/device cross-wiring (composes §11.4.111/.119/.176). Gate CM-COVENANT-114-178-PROPAGATION.
- **Corruption-isolated parallel git streams.** Constitution §11.4.179. Parallel git work streams requiring corruption-isolation are each an independent repo with its OWN .git (own object store/index/lock namespace), NOT git worktree checkouts sharing one common .git (composes §11.4.58/.119/.167/§9.2). Gate CM-COVENANT-114-179-PROPAGATION.
- **Stale-lock auto-reap before commit/push.** Constitution §11.4.180. Every commit/push wrapper auto-reaps a git lock whose recorded holder PID is dead (or, absent a PID, stale past a defined threshold with no live holder) before acquiring its own lock — NEVER removes a lock whose holder is alive (composes §11.4.84/.88/§9.2/.179). Gate CM-COVENANT-114-180-PROPAGATION.
- **Consistent feature-branch naming.** Constitution §11.4.181. One feature/logical group of workable items maps to EXACTLY ONE canonical branch name used identically on the main repo AND every touched owned submodule, recorded once in the claim registry, never re-invented (composes §11.4.29/.93/.113/.167/.176). Gate CM-COVENANT-114-181-PROPAGATION.

- **Track+branch work-stream identity label.** Constitution §11.4.182. Every agent/subagent label and operator-facing work-stream reference starts with a (T<N>/<branch> - <alias>) prefix, deterministically derived (never guessed), mechanically enforced by a PreToolUse guard hook (composes §11.4.75/.109/.178). Gate CM-COVENANT-114-182-PROPAGATION.
- **Maximal multi-agent utilization + full-constitution application.** Constitution §11.4.183. Every track maximizes applicable multi-agent working approaches (subagent-driven dev, independent review agents, parallel background streams) and applies the ENTIRE constitution — nothing skipped — with zero false/faulty/unverified results anywhere (composes §11.4.20/.58/.70/.125/.142/.165). Gate CM-COVENANT-114-183-PROPAGATION.
- **SonarQube CLI + tooling mandate.** Constitution §11.4.184. The SonarQube scanner CLI (sonar-scanner) is installed AND durably PATH-discoverable via shell rc, plus the shared constitution/scripts/sonarqube/ tooling consumed by reference (never copied); proven GREEN by install-check exit 0, never assumed; rootless podman only (composes §11.4.28/.74/.161/.166). Gate CM-COVENANT-114-184-PROPAGATION.
- **Manual QA final confirmation mandate.** Constitution §11.4.185. No scope of work / release is considered fully completed until it has received MANUAL testing confirmation by the project's QA team as the FINAL step — every automated gate is necessary but NOT sufficient; the agent hands off and waits, never self-certifies the manual step (composes §11.4.40/.52/.108/.126). Gate CM-COVENANT-114-185-PROPAGATION.
- **Anti-divergence cross-document consistency gate.** Constitution §11.4.186. Any project maintaining more than one representation of the same tracked data enforces cross-document consistency as a deterministic PASS/FAIL/SKIP gate running BEFORE any export render, doc/DB sync verify, or doc-set commit — never an after-the-fact audit; five decidable check families + self-validated analyzer (composes §11.4.86/.93/.106/.107/.148). Gate CM-COVENANT-114-186-PROPAGATION.
- **Process/thread-limit (RLIMIT_NPROC) awareness.** Constitution §12.12. Heavy parallel subagent/multi-process work is bounded by the OS per-user process/thread limit (ulimit -u / RLIMIT_NPROC); check thread headroom before scaling parallelism and treat exhaustion as a §12 host-safety event that yields unconditionally — an orthogonal axis to the §12.6 memory ceiling (composes §12.6/.7/ §11.4.58/.101/.103/.122). Gate CM-COVENANT-12-12-PROPAGATION.

Project-specific agent rules

Allowed CLI tools

- go, gofmt, go vet (server + Go bricks)
- gradle (Android bricks), git, gh, glab
- redocly, kubeconform, mmdc, pandoc (spec/doc validation)

Project-specific workflow

- Commit + push to ALL four upstreams regularly; merge to main when a milestone is done.
- New reusable submodule repos are created PUBLIC on GitHub + GitLab under the HelixDevelopment / vasic-digital orgs (pre-authorized).
- docs/research/main_specs/additions/ files are authoritative input — always analyze + fold them in.

Inheritance verification

Before a build/merge the inheritance gate MUST pass: `bash tests/test_constitution_inheritance.sh` (gate + \$1.1 paired mutation via `constitution/meta_test_inheritance.sh`).